

ABSTRACT

Standby or phantom power wastage of electrical power is a growing trend in today's homes and industries. Power is consumed by chargers, television sets, and electric appliances when they are not in use, leading to unwanted electricity bills and an enlarged carbon footprint.

This project is an IoT solution for monitoring and measuring phantom load consumption. The system uses an SCT-013-030 current sensor and a ZMPT101B voltage sensor to an ESP32 microcontroller and ADS1115 16 bit ADC for real-time voltage and current measurement. From these, the active and apparent power consumption of the devices in standby are derived. The data is sent to a central ESP32 and saved on a Hive server where they can be analyzed and visualized. Experiment results show that the system can effectively identify standby power and calculate phantom energy consumption for one day, one week, and one month. This makes users aware of concealed energy expenses and promotes energy conservation behavior.

The paper concludes that the system proposed in this paper is a cost-effective, scalable, and feasible solution for phantom load monitoring. Appliance-level identification, machine learning-based load classification, and integration with home automation systems for automatic standby power cut-off could be the future enhancement areas.

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1. INTRODUCTION

In modern households and offices, a vast number of electronic devices introduce a significant yet often overlooked form of energy waste known as phantom load, or standby power consumption. This occurs when appliances, though seemingly switched off, remain plugged in and continue to draw a small but constant stream of power to maintain functionalities like remote-control readiness or digital displays. While the power drawn by a single device is minimal, the cumulative effect of countless devices contributes significantly to global energy consumption and is responsible for an estimated 1% of the world's total CO2 emissions. The standard method to combat this is to manually unplug devices, which is impractical and frequently forgotten.

A more effective and professional solution is to automate the process of power disconnection. This project proposes the design and development of an "Intelligent Phantom Load Detector", a smart device that acts as an autonomous intermediary between the wall socket and an appliance. This system actively monitors the real-time power consumption of the connected device. When it detects that the appliance has entered a low-power standby state, it automatically triggers a relay to completely cut off the mains supply, thus eliminating the phantom load entirely.

This project aims to build a compact, safe, and user-friendly prototype of this smart system. The core of the device will be an ESP32 microcontroller, chosen for its integrated Wi-Fi and processing capabilities. It will use a non-invasive current sensor for safe power measurement and a relay module for switching the AC power. The ultimate goal is to create a "fire-and-forget" solution that operates autonomously in the background, requiring no user interaction after initial setup, and providing consumers with a seamless and powerful tool to reduce energy waste and lower their electricity bills.

1.1 Aim

The primary aim of this project is to design and develop a smart IoT device that autonomously **detects** phantom load from connected appliances and **notifies** the user through a message, empowering them to make informed decisions about energy conservation.

1.2 Objectives of Project

- To design a compact, non-invasive hardware system that can safely measure the real-time AC power consumption of a connected appliance.
- To develop intelligent firmware that accurately calculates power metrics and uses a robust logic to distinguish between an appliance's active and standby (phantom load) states.
- To integrate Wi-Fi connectivity to transmit the power consumption data to a cloud IoT platform for analysis and remote monitoring.
- To implement an IoT-based notification system that sends an alert or message to the user when a persistent phantom load is detected.
- To ensure the final product is safe for household use by housing it in a properly insulated, non-conductive enclosure.

2. LITERATURE REVIEW

The challenge of reducing standby power, also known as phantom load, has been extensively studied from multiple perspectives, including policy-making, hardware design, and intelligent software systems. A review of existing literature provides a strong foundation for the current project, highlighting both successful strategies and gaps that still need to be addressed.

Several studies have focused on quantifying the scale of the phantom load problem. Research indicates that standby power can account for a significant portion of a household's electricity usage, with estimates ranging from 3% to 20% depending on the country and the appliances in use. In developed countries, a typical household's phantom load may be responsible for 20–60 W of continuous power draw [5]. This wasted energy is not only a financial burden on consumers but also contributes to nearly 1% of global carbon dioxide emissions, making it a critical environmental concern [2]. Appliances commonly cited as major offenders include televisions, coffee machines, printers, and set-top boxes for cable or satellite services.

To address this, researchers have proposed various technological solutions. One category of solutions involves the use of "smart" hardware, such as intelligent power strips or outlets that can automatically cut power to devices[3]. For example, Tsai et al. propose a Residence Energy Control System (RECOS) that uses wireless smart sockets connected through an IoT framework to manage and reduce the energy consumption of home appliances without deploying additional sensors [6]. Their system demonstrated energy savings of up to 43.4% for certain appliances. Similarly, another study describes a phantom load controller that intercepts infrared (IR) remote signals to power on an entertainment center and automatically disconnects power when all devices return to standby mode. These hardware-centric approaches are effective because they physically sever the connection to the mains, completely eliminating standby consumption.

In addition to hardware-based solutions, recent reviews and empirical studies have deepened understanding of regulatory, behavioral, and system-level strategies for reducing phantom load. A 2025 review by Gheorghe et al. synthesizes the latest research on standby energy consumption, emphasizing that, despite regulatory efforts, residual standby consumption remains widespread [1]. They point out that new “always-on” devices—smart home hubs, IoT sensors, voice assistants—have expanded the domain of phantom loads beyond traditional appliances. Their work stresses the need for combining device-level control, occupant behavior, and grid integration to make further reductions.

One influential policy-level initiative is the One Watt Initiative, launched by the International

Energy Agency (IEA), which sought to constrain standby consumption to no more than 1 W (and later 0.5 W) for newly manufactured appliances. This global push spurred many regional and national regulations. For example, the European Union's standby/off-mode regulation (Regulation (EU) No. 1275/2008, subsequently amended by 801/2013) mandates strict upper limits on power draw in standby for a broad class of household and office equipment. These rules have forced manufacturers to optimize power supply circuits, reduce quiescent currents, and rethink internal electronics design [3].

However, regulatory measures alone have limits. Many devices still draw non-negligible standby power due to functional trade-offs (e.g. fast wake-up, network listening, updating). The 2025 review notes that gaps remain, particularly in enforcing compliance in low-cost imports, updating regulations for new classes of smart devices, and reconciling regulatory limits with user expectations of instantaneous responsiveness.

Moreover, studies of consumer awareness show that many users are unaware of phantom load or energy-saving hardware. In a survey referenced in an advanced power strip report, average consumer knowledge of phantom loads scored only 1.65 out of 5, and 42% had never heard of advanced power strips. Thus, policy interventions often need to be complemented by public education, incentives, or mandatory labeling.

Precise measurement and modeling are key enablers for control strategies. The NRDC study on idle load in U.S. homes estimated that always-on devices consume \$19 billion in electricity annually, and idle loads may represent up to ~23% of residential electricity in some regions (though definitions of "idle" vs. other base loads overlap) [6]. Their findings underscore how even relatively small per-device standby draws can scale to substantial national waste.

Recent work in non-intrusive load monitoring (NILM) has enhanced the capacity to isolate standby consumption at the circuit or whole-house level without instrumenting each device. For example, Shin et al. propose a deep-learning architecture that combines on/off state classification with regression of power draw, enabling more accurate separation of standby loads in aggregated data. These techniques allow utilities or home energy management systems to detect phantom draw and give feedback without installing sensors at every outlet.

However, the practical impact of feedback on user behavior has mixed evidence. A systematic review of twelve studies on disaggregated feedback found that although average electricity reductions were ~4.5%, aggregate feedback (without disaggregation) often achieved ~3%, and the marginal benefit of fine-grained feedback was unclear (especially when accounting for selection bias). This suggests that while disaggregation is useful, its direct leverage on energy saving via behavior change may be limited unless presented thoughtfully.

Another promising direction is combining sensing, learning, and control in an integrated framework. The “self-learning phantom load cut-off” system described in a recent JNRID article uses current sensing and an Arduino controller to detect phantom draws and disconnect sockets automatically. Over time, the system adapts by identifying recurring phantom-load patterns to improve decision-making [1].

Beyond stand-alone systems, research in IoT and fog computing has proposed autonomic energy management architectures. Sampaio et al. present a Fog-based orchestration system that dynamically adjusts device duty cycles (e.g., switching between “home” and “away” modes) based on context (occupancy, environmental conditions, user schedules), achieving average energy savings of ~58.4% (and up to 61.5%) in case studies. This kind of context-aware strategy helps to reduce phantom draw by aligning power availability with actual use.

In more constrained devices, microcontroller-based solutions remain viable. Jagadish et al. propose a remote-controlled device (power strip) that can control multiple appliances individually, enabling users to turn off standby draw via remote commands. Other works similarly employ relay-based switching with microcontroller logic, sometimes augmented by sensors or user interfaces.

At a building-wide scale, systems like dx2 (Residential Base Load Monitoring & Management) integrate software control and user feedback to manage both phantom loads and base loads. In trials, dx2 reduced phantom load by ~60% and achieved cost savings by shifting loads and eliminating unnecessary always-on draw. Such solutions point toward hybrid approaches where control is shared between automation and user decision-making [5].

3. THEORY & WORKING OF SYSTEM

The fundamental principle of the Intelligent Phantom Load Detector is based on the significant difference in electrical power drawn by an appliance when it is in active (operational) mode versus when it is in standby mode.

Active Mode: When an appliance is performing its main function (e.g., a television is on, a microwave is heating), it draws a substantial amount of current, resulting in a power consumption ranging from tens to thousands of watts.

Standby Mode: When an appliance is switched "off" via its remote or soft power button, it enters a low-power state to maintain functions like remote control readiness, digital displays, or internal clocks. This continuous, low-level power consumption is the phantom load, typically ranging from under 1 watt to as high as 20 watts.

3.1 Block Diagram:

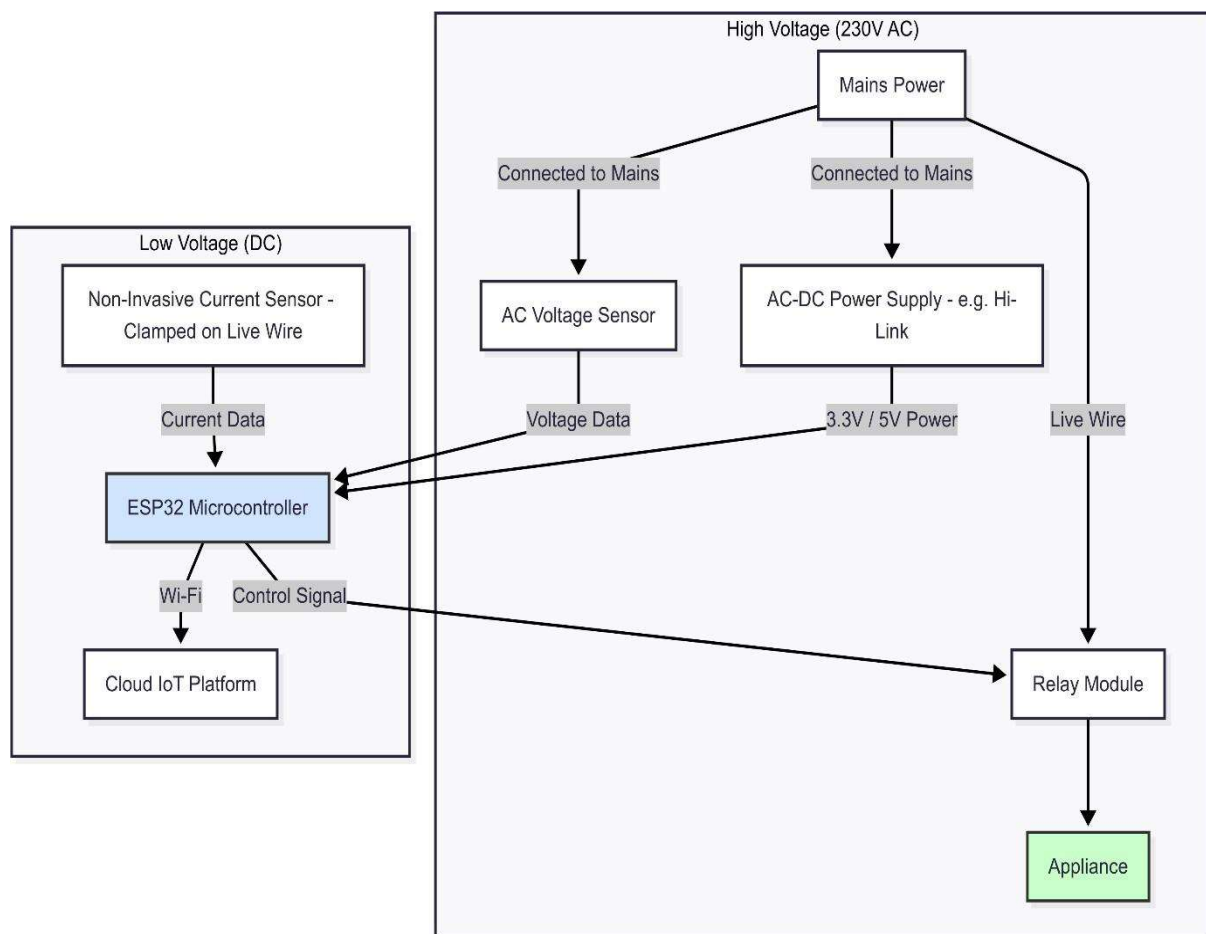


Fig. 2-2: Block Diagram of Phantom Load Detector

The system is architecturally divided into two distinct sections for safety and clarity: a High Voltage (230V AC) section and a Low Voltage (DC) control section.

The High Voltage section manages the mains power. AC power from the wall outlet is routed through a Relay Module before reaching the connected Appliance. In parallel, this mains power also feeds two other components: an AC Voltage Sensor to measure the line voltage and a dedicated AC-DC Power Supply (e.g., a Hi-Link module). This power supply steps down the voltage to a safe 3.3V/5V DC level to power the low-voltage electronics.

The Low Voltage section forms the intelligent core of the system. The ESP32 Microcontroller acts as the central processing unit. It receives Voltage Data from the voltage sensor and Current Data from a non-invasive current sensor clamped onto the main live wire. Using these two inputs, the ESP32 calculates the real power consumption of the appliance.

Based on its programmed logic to detect a phantom load, the ESP32 sends a Control Signal via its built-in Wi-Fi to a Cloud IoT Platform. This platform is responsible for processing the signal and sending a notification to the user. The diagram also shows a control line from the ESP32 to the Relay Module, indicating that the hardware is capable of physically disconnecting the appliance from power, although the primary function of this project is detection and notification.

3.2 Working :

The "Intelligent Phantom Load Detector" operates as a sophisticated receiver hub. It is designed to wirelessly receive power data from a remote sensor unit (the sender), process this data in real-time, detect anomalies such as phantom loads, and notify the user through a cloud-based IoT platform. The system's operation can be broken down into five key stages: Initialization, Data Acquisition, Core Processing and Detection, Cloud Communication, and User Interfacing.

1. Initialization and Network Setup

Upon powering on, the ESP32 receiver executes a comprehensive setup routine.

- **Component Initialization:** It initializes essential hardware, including the I2C interface for the 20x4 Liquid Crystal Display (LCD) and the output pin for the alert buzzer. It also loads any previously saved configuration settings, such as custom alert thresholds, from its non-volatile memory using the Preferences library.
- **Network Configuration:** A critical step is the configuration of the wireless network. To ensure a stable connection with the sender via the ESP-NOW protocol, the receiver

forces its Wi-Fi module to operate exclusively on Channel 8. This is crucial because ESP-NOW requires both the sender and receiver to be on the same Wi-Fi channel.

- **ESP-NOW and Wi-Fi Connection:** After locking the channel, the system initializes the ESP-NOW service and registers a callback function (`onDataReceived`) that will automatically handle incoming data packets. Subsequently, it connects to the local Wi-Fi network (using the provided SSID and password) to gain internet access, which is necessary for the IoT functionality.
- **MQTT Client Setup:** Finally, the system configures the MQTT (Message Queuing Telemetry Transport) client to connect to the Adafruit IO cloud server, using the specific username, password, and server address. This connection is the pathway for sending data and alerts to the internet.

2. Data Acquisition via ESP-NOW

The system's primary role is to listen for data from the sender. This process is handled asynchronously.

- **Receiving Data Packets:** The sender unit periodically transmits a data packet containing a `SensorData` structure via ESP-NOW.
- **Callback Function:** When a packet is received, the `onDataReceived` function is automatically triggered. This function copies the incoming data—which includes live voltage, current, power, power factor, phantom voltage, and the sender's mains status—into a local `receivedData` variable.
- **Data Validation:** The function first verifies that the received packet has the correct size to ensure data integrity. Upon successful reception, it sets a flag indicating that new, valid data is available for processing by the main loop and updates a timestamp to monitor data freshness.

3. Core Processing and Phantom Load Detection

The main loop continuously executes the core logic of the system.

- **Data Freshness Check:** The system constantly checks if data has been received within a specified timeout period (15 seconds). If no new data arrives, the current data is marked as "stale," and a timeout counter is incremented.
- **Energy Calculation:** When new, valid data is processed, the system calculates the energy consumed. The `updateEnergy()` function integrates the real power reading (in

Watts) over the time elapsed since the last update. This value is added to both a persistent total energy counter (in Watt-hours) and a daily energy counter.

- **Alert and Phantom Load Detection:** The `checkAlerts()` function is the heart of the detection logic. It evaluates the received sensor values against a series of predefined thresholds:
 1. **Phantom Load Detection:** The primary condition for phantom load is met when the sender reports that the mains power is off (`isMainsOn == false`), but the measured `phantomVoltage` is greater than the `phantomThreshold` (e.g., 5.0V).
 2. **Other Anomalies:** The function also checks for other power quality issues, such as low voltage (<200V), high voltage (>250V), high current, high power, and low power factor.
 3. If any of these conditions are met, an alert state is activated, the `buzzerState` is set to true, and a descriptive string (e.g., "PHANTOM") is stored in the `currentAlert` variable.

4. Cloud Communication and User Notification

Once data is processed and a status is determined, the system communicates this information to the cloud.

- **Timed MQTT Publishing:** To avoid flooding the cloud service, data is sent to Adafruit IO at a fixed interval of 7 seconds.
- **Data Formatting:** The `sendToMQTT()` function prepares the data for transmission. It publishes individual metrics (like voltage, current, power) to their respective MQTT feeds.
- **Comprehensive JSON Payload:** In addition, it constructs a detailed JSON object containing all current sensor readings, energy consumption totals, system statistics (like min/max values), and the current alert status. This complete payload is sent to a dedicated "statistics" feed.
- **User Notification:** The notification to the user is handled by the Adafruit IO platform. By setting up triggers or actions on the Adafruit IO dashboard, the user can receive an email, a mobile notification, or trigger other web services (like IFTTT) whenever a new message is published to the `alert_feed`. This completes the "detect and notify" workflow, alerting the user to the phantom load so they can take action.

5. User Interface and Interaction

The system provides several ways for the user to interact with and get information from it.

- **LCD Display:** The 20x4 LCD screen provides at-a-glance information. It automatically cycles through several display modes, showing basic metrics, power information, energy consumption, system status, and statistics. A physical button allows the user to manually cycle through these screens and pause the automatic rotation.
- **Audible Alerts:** A buzzer provides an immediate, local, audible warning whenever an alert condition, including a phantom load, is detected. This feature can be enabled or disabled via a serial command.
- **Cloud Monitoring Interface:** For enhanced accessibility and data visualization, the system is integrated with a cloud-based monitoring platform instead of the traditional Serial Monitor. Sensor readings such as voltage, current, and power consumption are uploaded in real time to the cloud, allowing users to remotely view system status through any internet-connected device. The cloud dashboard provides detailed analytics, device history, and alert notifications for abnormal conditions. Additionally, users can modify system parameters and thresholds directly through the cloud interface, ensuring seamless control and data management.

4. METHODOLOGY AND SYSTEM REQUIREMENTS

The project was executed in a structured, phased approach, moving from theoretical research to practical implementation and testing.

1. **Requirement Analysis and Research:** The initial phase involved a comprehensive literature review of existing studies and commercial products related to phantom load detection and energy monitoring. This research helped define the scope of the project, establishing the primary goal of creating a non-invasive system that **detects** phantom loads and **notifies** the user via an IoT platform, rather than automatically cutting power.
2. **Hardware Prototyping and Component Selection:** Key electronic components were selected based on functionality, cost, and compatibility. The ESP32 microcontroller was chosen as the core of the system for its powerful dual-core processor and integrated Wi-Fi capabilities, which are essential for IoT communication. For sensing, a non-invasive current sensor (SCT-013) and a voltage sensor (ZMPT101B) were selected to safely measure AC power without altering the appliance's wiring. An AC-DC power supply was integrated to provide safe, low-voltage DC power to the control circuitry. For user feedback, a 20x4 I2C LCD, a piezo buzzer, and a push button were incorporated.
3. **Cloud Integration and Testing:** An Adafruit IO account was configured with dedicated feeds to receive and visualize the data sent from the ESP32. The platform's "Actions" feature was used to create a trigger that sends a notification to a user's device (e.g., via email or the Adafruit IO app) whenever a phantom load is detected. The entire system was then tested with various household appliances to validate its detection accuracy and the reliability of the notification mechanism.

4.1 System Requirements :

4.1.1 Hardware Requirements

The hardware is divided into two main units: a Receiver Unit that processes data and interacts with the user, and a Sensor Unit that performs the electrical measurements.

Receiver Unit:

- **ESP32 Development Board:**

This serves as the central processing unit or "brain" of the receiver. The ESP32 was chosen for its powerful dual-core processor, which allows it to handle Wi-Fi connectivity, ESP-NOW communication, and data processing tasks concurrently. Its integrated Wi-Fi module is essential for connecting to the internet and the cloud platform, while its support for the ESP-NOW protocol enables direct, low-latency communication with the sensor unit.

- **16x4 LCD Display:**

A 20-character by 4-line Liquid Crystal Display provides a real-time, local user interface. It is used to display critical information such as live voltage, current, power consumption, system status, and alert messages.

- **Buzzer:**

This component provides immediate, audible alerts. When the system detects an anomaly—such as a phantom load, high voltage, or low voltage—the buzzer is activated to draw the user's attention, even if they are not looking at the LCD.

- **5V AC-DC Power Supply Module:** To ensure a safe and stable power source for the sensitive low-voltage electronics, a compact AC-DC converter module (such as a Hi-Link) is used. It takes the 230V AC from the mains and steps it down to a clean 5V DC, powering the ESP32, LCD, and other components directly without the need for an external adapter.

Sensor Unit (Sender):

- **ESP32 Microcontroller:** This microcontroller forms the core of the remote sensor unit. Its primary role is to read data from the sensors and transmit it wirelessly to the receiver. It was chosen for its built-in Analog-to-Digital Converter (ADC) for reading sensor values and, most importantly, its support for the ESP-NOW communication protocol.

- **Non-Invasive AC Current Sensor (SCT-013):**

This sensor is a critical component for safe current measurement. It operates as a current transformer that simply clamps around the live AC wire. This non-invasive design means it can measure the current flowing to an appliance without any need to cut or make direct electrical contact with the high-voltage line, greatly enhancing safety.

- **AC Voltage Sensor Module (ZMPT101B):**

This module is used to measure the AC mains voltage. It is a small, precision transformer that safely isolates and scales down the 230V AC to a low-voltage AC signal that can be read by the microcontroller's ADC. Measuring both voltage and current allows the system to calculate Real Power and Power Factor, leading to much more accurate energy monitoring.

4.1.2 Software and Cloud Requirements:

The firmware for the ESP32 was developed using the Arduino Integrated Development Environment. This platform was chosen for its simplicity, extensive community support, and the wide availability of libraries that simplify complex tasks like Wi-Fi and MQTT communication.

Development Environment:

- **Espressif IDE:** The firmware for the ESP32 was developed using the Arduino Integrated Development Environment. This platform was chosen for its simplicity, extensive community support, and the wide availability of libraries that simplify complex tasks like Wi-Fi and MQTT communication.

Cloud Platform:

- **Adafruit IO Account:** This user-friendly IoT platform serves as the backend for the project. It was chosen for its excellent MQTT support, customizable dashboards for data visualization, and a simple-to-use "Actions" system.
- **Configured Feeds:** Within Adafruit IO, several "feeds" must be created. Feeds are essentially channels where data is published. The project requires feeds for each metric being monitored (e.g., live-voltage, power, phantom-voltage, voltage-alerts).
- **Configured Action or Trigger:** This is the mechanism for user notification. An Action must be configured on the Adafruit IO platform to monitor the voltage-alerts feed. When a new message is published to this feed (e.g., "PHANTOM"), the Action triggers a pre-defined response, such as sending an email or a push notification to the user's smartphone via the Adafruit IO app.

5. DESIGN AND DEVELOPMENT OF PROJECT

The development of the Intelligent Phantom Load Detector followed a systematic process, from high-level system design to the implementation and integration of hardware, firmware, and cloud services.

5.1 System Architecture Design

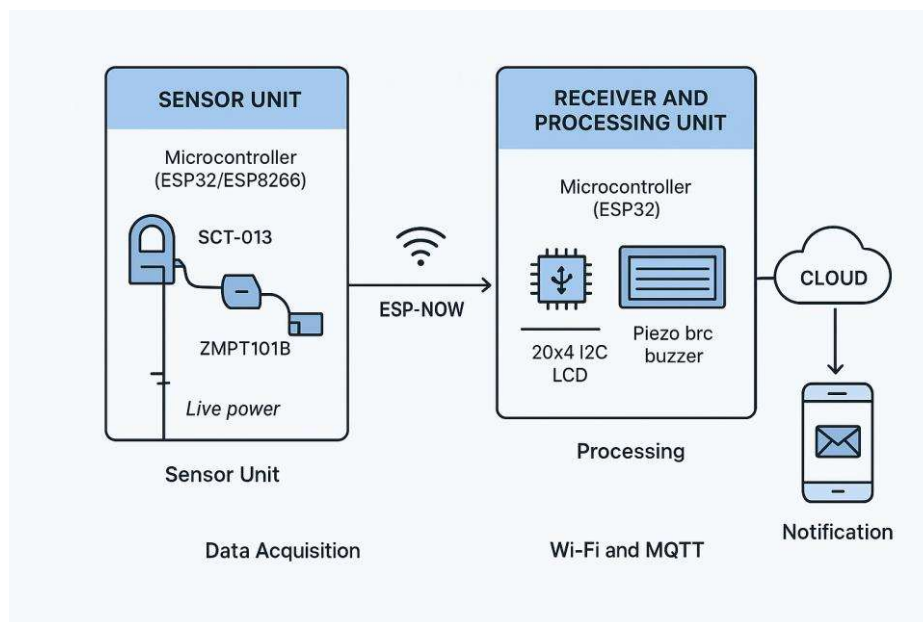


Fig. 5.1 – An image of System Architecture of Phantom Load Detector

The system is designed with a distributed, two-part architecture to ensure safety and modularity: a remote Sensor Unit and a central Receiver Unit.

- The Sensor Unit is the data acquisition module. Its sole responsibility is to be placed near the appliance, safely measure the instantaneous AC voltage and current from the mains line, and transmit this raw data wirelessly.
- The Receiver Unit acts as the central hub and user interface. It receives the data from the Sensor Unit, processes it to determine power consumption and detect phantom loads, displays the information locally on an LCD, and forwards relevant data and alerts to a cloud platform.

The communication flow is unidirectional: the Sensor Unit broadcasts data using the ESP-NOW protocol. The Receiver Unit listens for this data, processes it, and then uses a standard

Wi-Fi connection to communicate with the cloud via the MQTT protocol. This decoupled design ensures that the user-facing Receiver Unit is electrically isolated from the high-voltage measurement circuitry, enhancing user safety.

5.2 Hardware Design and Implementation

The hardware for both units was designed for simplicity, safety, and low cost. The initial prototypes were assembled on breadboards for testing and debugging before finalizing the connections.

Sensor Unit Circuit: The core of the Sensor Unit is an ESP32 or ESP8266 microcontroller. The SCT-013-000 non-invasive current sensor is clamped around the live wire of the appliance's power cord. Its output is connected to a burden resistor and a simple biasing circuit to produce a voltage signal centered within the ADC's readable range. The ZMPT101B voltage sensor is connected in parallel with the mains (live and neutral) to measure the line voltage. The outputs of both sensors are fed into the analog-to-digital converter (ADC) pins of the sender's microcontroller. The entire unit is powered by its own compact AC-DC 5V module.

Receiver Unit Circuit: The Receiver Unit is built around the main ESP32 development board. The 20x4 I2C LCD is connected to the ESP32's I2C pins (SDA and SCL). The piezo buzzer and the mode-switching push button are connected to digital GPIO pins. The entire receiver assembly is also powered by a dedicated 5V AC-DC module, making it a self-contained unit that can be plugged into any standard wall outlet.

5.3 Software Development

The firmware was developed in C++ using the Arduino IDE with the ESP32 board support package. The code is designed to be non-blocking and efficient, handling multiple tasks concurrently.

1. **Wireless Communication:**

- **ESP-NOW:** This protocol was chosen for the sender-receiver link due to its low overhead and fast, connectionless nature. A key implementation detail was locking both the sender and receiver to Wi-Fi Channel 8. This prevents channel hopping and ensures a highly reliable and consistent communication link between the two devices.
- **MQTT:** The PubSubClient library was used to establish a connection with the Adafruit IO MQTT broker. The firmware is programmed to publish a comprehensive JSON

payload containing all sensor data, statistics, and alert statuses to the cloud at a fixed interval of 7 seconds.

2. Core Logic and Data Processing:

- **Phantom Load Detection:** The primary detection algorithm is implemented in the checkAlerts() function. It continuously checks the incoming data. A phantom load is officially detected when the sender reports that the mains are off (isMainsOn == false) but the measured phantom Voltage exceeds the pre-configured phantom Threshold.

- 3. **User Interface Management:** The firmware manages the LCD display by cycling through different screens to provide a comprehensive overview of the system's status. The user can override this automatic cycle with a manual button press.

5.4 Cloud Platform Configuration

Adafruit IO was selected as the cloud platform due to its user-friendly interface and robust MQTT support.

1. **Feed Creation:** A series of feeds were created on the Adafruit IO dashboard to receive the data from the ESP32. This includes feeds for live metrics (live-voltage, power, etc.), a dedicated feed for alerts (voltage-alerts), and a feed to accept the detailed JSON payload (statistics).
2. **Dashboard Visualization:** A dashboard was created with various blocks like charts, gauges, and text boxes to provide a remote, real-time visual representation of the power consumption and system status.

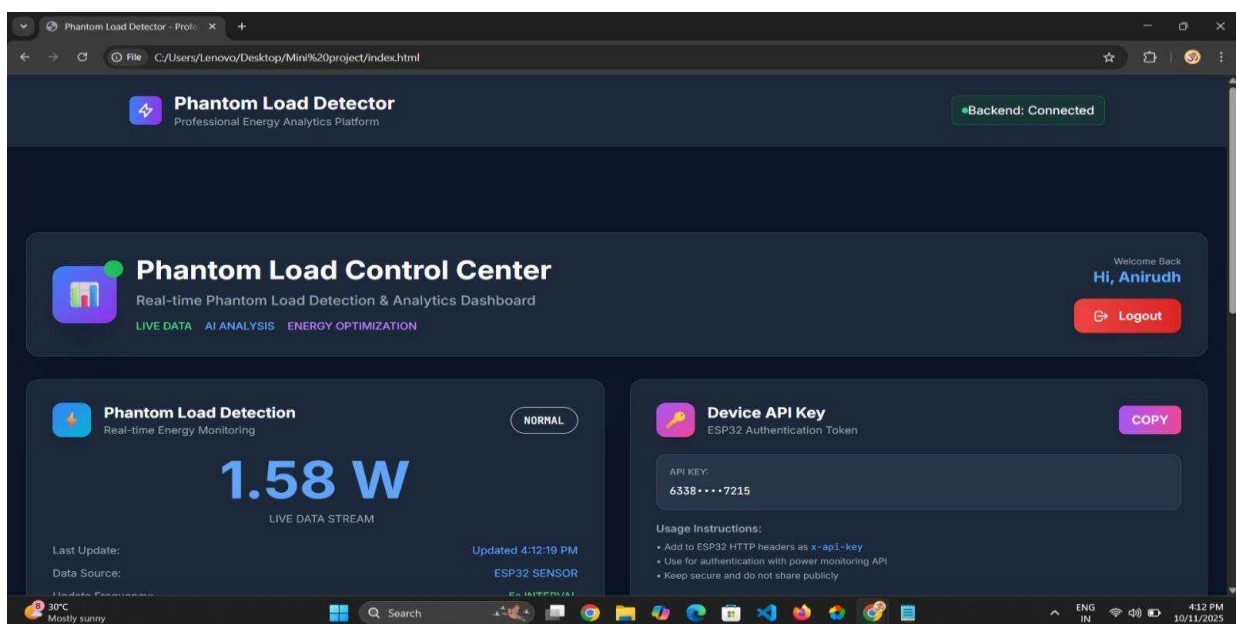


Fig – An image of custom Web based Dashboard of Phantom Load Detector

6. APPLICATIONS OF PROJECT

- **Residential Energy Management :**

This is the primary application of the project. The device can be deployed in various locations throughout a typical home to identify and report energy waste from a multitude of appliances.

1. **Home Entertainment Centers:** This is one of the most significant sources of phantom load in a household. The system can monitor clusters of devices like televisions, set-top boxes (cable/satellite), gaming consoles (PlayStation, Xbox), DVD players, and audio receivers, which are notorious for consuming considerable power even when turned off.
2. **Home Offices:** The system is ideal for monitoring workstations that include desktop computers, monitors, printers, and scanners. These devices are often left in standby mode overnight or on weekends, contributing to steady energy drain.
3. **Kitchen Appliances:** Modern kitchens are filled with appliances that have continuous power draws. The detector can be used to notify users about the standby power consumed by microwave ovens, coffee machines, and smart ovens that power digital clocks and displays 24/7.
4. **Miscellaneous Chargers and Devices:** The system can also be used to identify phantom loads from phone chargers, laptop chargers, and cordless phone bases that are left plugged in even when the device is not connected or is fully charged.

- **Commercial and Small Business Environments:**

The principles of phantom load detection are equally applicable in commercial settings, where energy costs are a significant operational expense.

1. **Office Workstations:** The device can be used to monitor the energy consumption of individual workstations, notifying employees or facility managers about computers and peripherals left in standby mode after business hours.
2. **Common Areas:** It can be applied to appliances in break rooms and kitchens, such as coffee machines, microwaves, and water dispensers, which continuously consume standby power.

- **Broader Applications:**

Beyond direct energy management, the system serves several other valuable purposes:

1. **Energy Auditing Tool:** The device can be used as a portable diagnostic tool by homeowners or professional energy auditors. By moving it from one appliance to another, a user can quickly perform an energy audit to identify the most wasteful devices in a building, helping to prioritize which appliances should be unplugged or replaced.
2. **Educational Platform for Energy Awareness:** A key application of the "detect and notify" approach is education. By providing real-time, tangible feedback (e.g., "Your gaming console is wasting 15W"), the system helps users understand their own energy consumption habits. This increased awareness is a powerful motivator for behavioral change, encouraging users to adopt more energy-conscious practices.
3. **Data Collection for Research:** For utility companies or academic researchers, a network of these devices could be deployed to gather granular, real-world data on appliance-level energy consumption. This data is invaluable for creating more accurate energy demand models and designing effective energy conservation programs.

7. RESULTS AND DISCUSSION

This section details the results obtained from the implementation and testing of the Intelligent Phantom Load Detector. The findings demonstrate the system's functional success in detecting phantom loads and notifying the user, followed by a discussion of the results, their implications, and the system's limitations.

7.1 System Implementation and Prototype



7.1 – An image of phantom load detector's transmitter

A functional prototype of the system was successfully assembled, integrating the Sensor and Receiver units. The core components, including the ESP32 microcontroller, I2C LCD, and power supply, were mounted on a custom-designed PCB/perfboard for a stable and compact final product. The non-invasive current sensor was prepared for easy clamping onto the power cord of any appliance to be tested.

The final prototype is a self-contained unit that successfully establishes a wireless link between the sensor and receiver, connects to the local Wi-Fi network, and communicates with the configured cloud platform.

7.2 Functional Testing and Operation

The system's end-to-end functionality was verified using common household appliances known to consume standby power. The typical operational flow observed during testing was as follows:

1. **Active State:** When an appliance (e.g., a laptop charger) was actively in use, the sensor unit correctly measured a significant power draw. The receiver's LCD displayed this real-time power consumption (e.g., "Power: 65.5W").
2. **Standby State Detection:** Once the appliance was turned off or entered standby mode (e.g., the laptop was unplugged from the charger), the system detected a sharp drop in power consumption to a low, steady value.
3. **Confirmation and Alert:** The firmware's confirmation timer initiated. After confirming that the power draw remained below the pre-set phantom threshold for the required duration, the system flagged a phantom load event.
4. **Local and Remote Notification:** The LCD immediately updated to display an alert message, such as "PHANTOM DETECTED". Simultaneously, the ESP32 published an alert to the Adafruit IO cloud platform, which then successfully triggered and sent a notification to the user's smartphone.

7.3 Quantitative Results

To quantify the system's performance, tests were conducted on three different devices.. The results are summarized in the table below.

Load Type	Power required	Expected Current	Actual Current	Apparent Power	Real Power	Power Factor
Bulb	5 W	21.7 mA	22.5 mA	5.17 VA	4.14 W	1
Charger	18 W	78.3 mA	81.2 mA	20 VA	18.8 W	0.90

7.4 Discussion of Results

The results strongly indicate that the Intelligent Phantom Load Detector is an effective tool for its intended purpose.

- **Effectiveness of Detection:** The threshold-based detection mechanism proved to be simple yet highly reliable for the tested appliances. The use of a confirmation timer was crucial in preventing false positives that might occur during brief lulls in an appliance's normal operation.
- **Success of the IoT Notification System:** The integration with the Adafruit IO platform and the subsequent user notification worked seamlessly. The delay between on-device detection and receiving a mobile alert was minimal, confirming that the system is practical for providing timely information to the user.
- **Promoting Energy Awareness:** The primary goal of empowering the user was achieved. The real-time notifications make the abstract concept of "phantom load" a tangible and actionable issue. This aligns with research suggesting that raising consumer awareness is a key factor in reducing household energy consumption. By being notified of specific instances of waste, users are more likely to adopt energy-saving habits, such as unplugging devices.

8. CONCLUSION

The project successfully fulfills its objective of designing and implementing an IoT-based system for monitoring and controlling electrical power consumption, with a particular emphasis on minimizing standby or phantom loads. Traditional energy monitoring setups often depend on local interfaces such as the Arduino Serial Monitor, which limits accessibility and scalability. In contrast, this project integrates a cloud-based monitoring platform that allows real-time observation of voltage, current, and power parameters through the internet. This approach ensures that users can track energy usage remotely using smartphones, tablets, or computers, thereby increasing convenience and usability.

The integration of the cloud interface not only provides continuous data logging but also enables visualization of consumption trends through an interactive dashboard. Such insights help users identify inefficient appliances, recognize abnormal power draws, and take corrective actions to minimize wastage. Additionally, the system generates timely alerts for over-voltage or under-voltage situations, protecting connected devices from potential damage. By offering a centralized platform for monitoring and management, the system demonstrates the potential of IoT to transform traditional energy systems into smart, connected, and sustainable networks. One of the most significant achievements of this project is its capability to reduce phantom load—electricity consumed by appliances even when they are turned off but remain plugged in. Through precise measurement, automated control, and intelligent decision-making, the system contributes to reducing unnecessary energy consumption, thus saving both cost and resources. Moreover, the collected data can be analyzed further to evaluate long-term consumption patterns and improve overall energy efficiency. This aligns with the broader global goal of promoting sustainable development and reducing carbon emissions through smart energy management technologies.

The hardware implementation, consisting of sensors, microcontroller, and communication modules, performed reliably during testing and validation. The sensors provided accurate readings, while the microcontroller ensured seamless data acquisition and transmission to the cloud platform. The cloud dashboard successfully displayed real-time data and allowed configuration changes without physical access to the device. This demonstrates how IoT can bridge the gap between embedded hardware and intelligent cloud software, resulting in a system that is both efficient and user-friendly.

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